From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ANSWERS, HINTS & SOLUTIONS
FULL TEST – II
(Main)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	A	A	B
2.	B	C	A
3.	C	B	A
4.	B	D	A
5.	D	B	B
6.	C	C	B
7.	A	C	B
8.	C	C	C
9.	A.	B	B
10.	C	B	B
11.	D	D	D
12.	C	B	B
13.	C	C	A
14.	A	B	A
15.	B	C	B
16.	D	B	A
17.	A	C	A
18.	C	B	A
19.	A	C	A
20.	C	B	C
21.	A	B	D
22.	D	B	C
23.	A	D	D
24.	C	B	B
25.	D	D	C
26.	B	B	C
27.	B	D	C
28.	D	D	A
29.	C	C	A
30.	D	D	A

Physics

PART – I

1. Avg. speed $3 = \dfrac{\sqrt{(v_r t)^2 + (v_{mr} t)^2}}{t}$

$\Rightarrow v_r^2 + 5 = 9$

$\Rightarrow v_r = 2 m/s$

2. $v_{avg} = \dfrac{\frac{1}{2} \times \frac{t}{2} \times v + \frac{1}{2} \times v}{t} = \dfrac{3v}{4}$

3. $\dfrac{m}{2} g - T = \dfrac{m}{2} a$ …(i)

$T \cos 60° = \dfrac{ma}{\cos 60°}$ …(ii)

Solving (i) and (ii) acceleration of ring = $\dfrac{2g}{9}$

4. Work done by all the forces on the block equal to change in kinetic energy.

6. No effect of 'a' and 'g' on time period of spring pendulum.

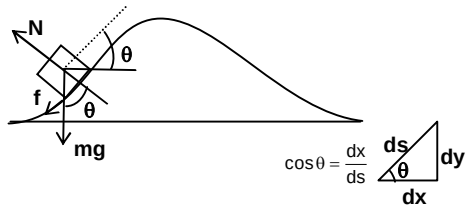
7. $\dfrac{1}{\lambda_{3-1}} = R\left(\dfrac{1}{1^2} - \dfrac{1}{3^2}\right) = \dfrac{8R}{9}$

$\dfrac{1}{\lambda_{2-1}} = R\left(\dfrac{1}{1^2} - \dfrac{1}{2^2}\right) = \dfrac{3R}{4}$

$\Rightarrow \dfrac{\lambda_{3-1}}{\lambda_{2-1}} = \dfrac{27}{32}$

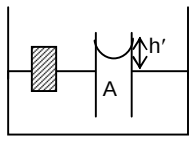
8. Work done by friction $= \int \vec{F} \cdot \vec{ds} = \int_0^x \mu mg \cos\theta \dfrac{dx}{\cos\theta}$

$= \mu mg\, x = 20$ J



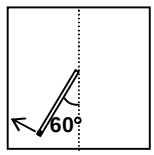
9. There will be no change.

$\therefore \quad h' = h$



10. Conservation of energy $\dfrac{1}{2} m v^2 - mg\dfrac{\ell}{2} = mg\dfrac{\ell}{2} + m\sqrt{3}\left(2\ell\dfrac{\sqrt{3}}{2}\right) g$

$v = \sqrt{8g\ell}$



11. Potential across capacitor is zero, hence energy stored is zero.

12. $v_A - v_B = \int E.\, dx = 10\ (1)$
$= 10$ volt.

13. $Z = \sqrt{R^2 + X^2} = \sqrt{9 + X^2}$
but $\cos\phi = \frac{R}{Z} = \frac{3}{5}$
$X = 4\ \Omega$.

14. $\varepsilon = \frac{\sigma}{\varepsilon_0}$
$v_A - v_B = \frac{\sigma}{\varepsilon_0}(b) - \frac{\sigma}{\varepsilon_0}(a)$
$\therefore\ W = Q(v_A - v_B) = \frac{Q\sigma}{\varepsilon_0}(b - a)$

15. $\omega = 0 + 1 \times 10 = 10\ \text{rad/sec}^2$
$\therefore\ v = r\omega = 1 \times 10 = 10\ \text{m/s}$
$\bar{B} = \frac{\mu_0}{4\pi}\frac{q(\bar{v} \times \bar{r})}{r^3} \Rightarrow |\bar{B}| = \frac{\mu_0 qv}{4\pi r^2}$
$B = \frac{10^{-7} \times 0.1 \times 10}{(1)^2} = 10^{-7}\ T$

16. $P = \sigma A T^4$
$\Rightarrow \frac{P_A}{P_B} = \frac{A_B}{A_A}\left(\frac{T_B}{T_A}\right)^4 \Rightarrow 16 = \frac{2}{1}\left(\frac{T_B}{T_A}\right)^4 \Rightarrow T_B = T_A(8)^{1/4}$
Since $\lambda_m T$ = constant, $\frac{\lambda_A}{\lambda_B} = \frac{T_B}{T_A} = (8)^{1/4}$
$\Rightarrow \lambda_B = \frac{\lambda_A}{(8)^{1/4}} = \frac{5000}{(8)^{1/4}}$ Å

17. For adiabatic process, $TV^{\gamma-1}$ = constant
$T\left(\frac{m}{\rho}\right)^{\gamma-1}$ = constant
$\frac{T}{\rho^{\gamma-1}}$ = constant
$\rho \propto T^{1/(\gamma-1)} \Rightarrow \frac{1}{\gamma - 1} = 3 \Rightarrow \gamma = 4/3$
$f = \frac{2}{\gamma - 1} = \frac{2}{\left(\frac{4}{3} - 1\right)} = 6$

18. P(x, y), A(–3a, 0), B = (3a, 0)
$V_{PA} = \frac{1}{4\pi\varepsilon_0}\frac{-2Q}{PA}$, $V_{PB} = \frac{1}{4\pi\varepsilon_0}\frac{Q}{PB}$
According to equation

$$\frac{1}{4\pi\varepsilon_0}\left(\frac{-2Q}{PA}\right)+\frac{1}{4\pi\varepsilon_0}\frac{Q}{PB}=0$$

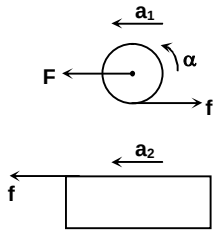
$$\frac{2}{PA}=\frac{1}{PB}\Rightarrow 4PB^2=PA^2$$

$$(x-5a)^2+y^2=(4a)^2$$

19. $i=\sqrt{5}A$

$$\frac{q_m^2}{2C}=\frac{q^2}{2C}+\frac{1}{2}Li^2\Rightarrow q_{max}=6\ C$$

21. $f=ma_2$



$$\alpha=\frac{\tau}{I}=\frac{2f}{mR}$$

$\Rightarrow a_2=F/4m,\ f=F/4$

22. The process is equivalent to $TV^{1/2}=C$

Compare with $TV^{x-1}=C\Rightarrow x=3/2$

$$\Rightarrow\ C=\frac{R}{\gamma-1}+\frac{R}{1-X}=\frac{R}{2/3}+\frac{R}{1-(3/2)}=\frac{3}{2}R-2R=-\frac{1}{2}R$$

23. Intensity will be highest at the nearest point.

28. $\tan\theta=\frac{u^2}{Rg}\Rightarrow R=\frac{u^2}{g\tan\theta}=\frac{100}{10\times\sqrt{3}}=\frac{10}{\sqrt{3}}$

Chemistry

PART – II

14. $E = \frac{hc}{\lambda} = 2.9 \times 10^{-19}$ J

Total energy of 10 quanta = $10 \times 2.9 \times 10^{-19}$J

Energy stored for process $= \frac{112 \times 4.18 \times 10^3}{6 \times 10^{23}} = 7.8 \times 10^{-19}$ J

% efficiency $= \frac{7.8 \times 10^{-19}}{29 \times 10^{-19}} \times 100$

= 26.9%

15. $\frac{r_a}{r_b} = \frac{2}{1} = \sqrt{\frac{M_b}{M_a}}$

$V_{rms} \propto \sqrt{\frac{T}{M}}$ (As $V_{rms} = \sqrt{\frac{3RT}{M}}$)

$\frac{V_{rms(a)}}{V_{rms(b)}} = \sqrt{\frac{T_a \times M_b}{T_b \times M_a}} = \frac{2}{1} \times \frac{\sqrt{2}}{1} = \frac{2\sqrt{2}}{1}$

16. $\ell n \frac{k_1'}{k_1} = \frac{E_1}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$ ... (i)

$\ell n \frac{k_2'}{k_2} = \frac{E_2}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$ ... (ii)

Eqn. (ii) – Eqn.(i)

$\ell n \frac{k_2' \times k_1}{k_2 \times k_1'} = \frac{(E_2 - E_1)}{R}\left[\frac{1}{T_1} - \frac{1}{T_2}\right]$

(For equimolar formation of B and C, $k_2' = k_1'$)

$\ell n\left(\frac{k_1}{k_2}\right) = \left(\frac{8314}{8.314}\right)\left(\frac{T_2 - 300}{300 \times T_2}\right)$

$\ell n2 = = \left(\frac{8314}{8.314}\right)\left(\frac{T_2 - 300}{300 \times T_2}\right)$

T_2 = 329.77 K.

17. $2SO_3(g) \rightleftharpoons 2SO_2(g) + O_2(g)$

at equilibrium $(1-2x)$ $(2x)$ x

Only SO_2 will oxidized.

Equivalent of SO_2 = Equivalent of $KMnO_4$

$2x \times 2 = 0.2 \times 5$

$2x = 0.5$

$K_c = \frac{\left[\frac{0.5}{2}\right]^2\left[\frac{0.25}{2}\right]}{\left[\frac{0.5}{2}\right]^2} = 0.125$

18. $\Delta T_b = i \times k_b \times m$

$2.08 = 0.52 \times 1 \times i$

$i = 4$

It means salt on dissociation gives 4 ions. Thus the salt that gives 4 ions is $K_3[Fe(CN)_6]$.

19. $Cu^{+2} + e^- \longrightarrow Cu^+$; $E^o = X_1V$ $\Delta G_1^o = -FX_1$ ... (i)

$In^{+3} + 2e^- \longrightarrow In^{+1}$ $E^o = X_2V$ $\Delta G_2^o = -2FX_2$ ...(ii)

$In^{+2} + e^- \longrightarrow In^+$ $E^o = X_3V$ $\Delta G_3^o = -FX_3$ ... (iii)

From Eqn. (i) + (iii) – (ii)

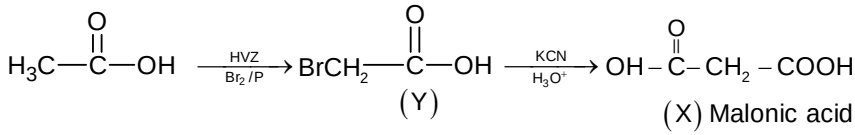
$In^{+2} + Cu^{2+} \longrightarrow In^{+3} + Cu^+$ $\Delta G^o = -FE^o$

$\Delta G^o = -F(X_1 + X_3 - 2X_2) = -FE^o$

$E^o = (X_1 + X_3 - 2X_2)$

20. Due to Fajan's rule.

23. Green ppt. is – $Cr(OH)_3$

25. 

26. 1° aromatic amine on diazotisation followed by coupling with β-napthol gives azo dye test.

28. (A) Viscosity α intermolecular attraction
α intermolecualr H-bonding tendency

(B) No intramoecular H-bonding in the cis-isomer due to loss in planarity as a result of steric and electronic repulsion.
$\therefore$ stability α symmetry.

(C) Even though o-nitrophenol shows intramolecular H-bonding but has greater – R and – I effect than –Cl.

(D) $H_3BO_3 + 3\ C_2H_5OH \rightarrow B(OC_2H_5)_3 + 3H_2O$.
$\therefore$ No cage lattice structure of H_3BO_3.

29. Because phenol is less acidic than H_2CO_3.

30. S is less electronegative and therefore more reactive towards S_N2.

Mathematics

PART – III

1. Given equation can be written as $|x-1|^2 + |x-1| - 6 = 0$
$t^2 + t - 6 = 0$
$\Rightarrow t = 2$ or $-3 \Rightarrow |x-1| = 2$
$\Rightarrow x - 1 = \pm 2$ or $x = 3, -1$

2. $|z_1 - z_2|^2 + |z_2 - z_3|^2 + |z_3 - z_4|^2 + |z_4 - z_1|^2$
$= 2(|z_1|^2 + |z_2|^2 + |z_3|^2 + |z_4|^2) + |z_1 + z_3|^2 + |z_2 + z_4|^2 \geq 8 + |z_1 + z_3|^2 + |z_2 + z_4|^2$

3. $T_r = \dfrac{r}{r^4+4} = \dfrac{r}{(r^2+2)^2 - 4r^2} = \dfrac{r}{(r^2+2r+2)(r^2-2r+2)}$

$= \dfrac{1}{4}\left[\dfrac{1}{r^2-2r+2} - \dfrac{1}{r^2+2r+2}\right] = \dfrac{1}{4}\left[\dfrac{1}{(r-1)^2+1} - \dfrac{1}{(r+1)^2+1}\right]$

$\Rightarrow S_n = \dfrac{1}{4}\left[1 + \dfrac{1}{2} - \dfrac{1}{n^2+1} - \dfrac{1}{(n+1)^2+1}\right]$

$\Rightarrow S_5 = \dfrac{1}{4}\left[1 + \dfrac{1}{2} - \dfrac{1}{26} - \dfrac{1}{37}\right] = \dfrac{1}{4}\left[\dfrac{38\times 37 - 26\times 1}{26\times 37}\right] = \dfrac{1}{4}\times\dfrac{1380}{26\times 37} = \dfrac{345}{26\times 37}$

$\Rightarrow 37S_5 - \dfrac{7}{26} = \dfrac{338}{26} = 13$

4. Number of terms = Number of non–negative integral solution of the equation $p + q + r + s = 50$
$= {}^{50+4-1}C_{50} = {}^{53}C_{50} = {}^{53}C_3$

5. If a function is symmetric about two mutually perpendicular lines, it must be symmetric about their point of intersection.

6. $-\dfrac{x^5}{\sqrt{1-x^6}} - \dfrac{y^5}{\sqrt{1-y^6}}\dfrac{dy}{dx} = a\left(x^2 - y^2\dfrac{dy}{dx}\right)$

$\Rightarrow \left(ay^2 - \dfrac{y^5}{\sqrt{1-y^6}}\right)\dfrac{dy}{dx} = ax^2 + \dfrac{x^5}{\sqrt{1-x^6}}$

$\Rightarrow \dfrac{dy}{dx} = \dfrac{\sqrt{1-y^6}}{\sqrt{1-x^6}} \times \dfrac{x^2\left(a\sqrt{1-x^6} + x^3\right)}{y^2\left(a\sqrt{1-y^6} - y^3\right)}$ ..... (1)

Also, $\dfrac{(1-x^6)-(1-y^6)}{\sqrt{1-x^6}-\sqrt{1-y^6}} = a\left(x^3 - y^3\right)$ or $x^3 + y^3 = a\left(\sqrt{1-y^6} - \sqrt{1-x^6}\right)$

$\Rightarrow a\sqrt{1-y^6} - y^3 = a\sqrt{1-x^6} + x^3$

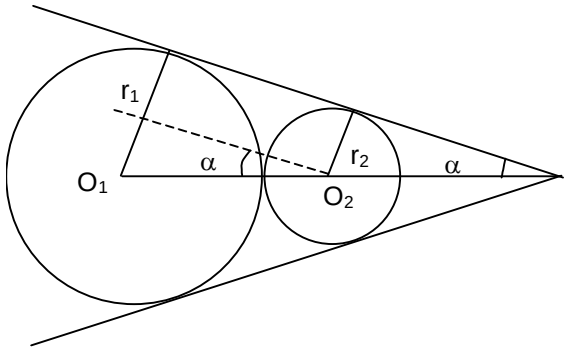
Hence, from equation (1), $\dfrac{dy}{dx} = \dfrac{\sqrt{1-y^6}}{\sqrt{1-x^6}} \times \dfrac{x^2}{y^2}$

7. For a triangle formed by joining any three co–normal points, orthocentre consides with the fourth point.

8. HM of SP and SQ = $\dfrac{2(3)(6)}{3+6} = 4$ = semi latus rectum.

9. If a circle cuts two circles orthogonally, radical axis of the two circles pass through the centre of the first circle
Here radical axis of given circles is $6x - 4y + 4 = 0$ or $3x - 2y + 2 = 0$
Hence $3\alpha - 2\beta + 2 = 0 \Rightarrow \frac{3}{4}\alpha - \frac{\beta}{2} = -\frac{1}{2}$

10. $\sin\alpha = \frac{r_1 - r_2}{r_1 + r_2}$
$\Rightarrow \theta = 2\sin^{-1}\left(\frac{r_1 - r_2}{r_1 + r_2}\right)$



11. Let $\vec{r} = a\vec{i} + b\vec{j} \Rightarrow a^2 + b^2 = 1$ ...(1)
Also,
$\cos 45° = \frac{\vec{r}\cdot(\vec{i}+\vec{j})}{|\vec{r}||\vec{i}+\vec{j}|} \Rightarrow a + b = 1$ ...(2)
$\cos 60° = \frac{\vec{r}\cdot(3\vec{i}-4\vec{j})}{|\vec{r}||3\vec{i}-4\vec{j}|} \Rightarrow 3a - 4b = \frac{5}{2}$ ...(3)
There exists no real values of a and b satisfying (1), (2) and (3)
Hence no such unit vector exists

12. $(1 + x)^n = 3 + \frac{8}{3} + \frac{80}{3^3} + \frac{240}{3^4} + ... = 1 + nx + \frac{n(n-1)x^2}{2} + ...$
On comparison, $n = -3$ and $x = -\frac{2}{3}$

13. From the given conditions, $\frac{y-x}{b-a} \neq \pm 1$, $\frac{z-y}{c-b} \neq \pm 1$, $\frac{z-x}{c-a} \neq \pm 1$
$\Rightarrow x + a \neq y + b \neq z + c$ The determinant is a symmetric one. The determinant will be equal to zero if $x + a + y + b + z + c = 0$
but $a + b + c = 0$ (given)
$\Rightarrow x + y + z = 0$
$\Rightarrow x + z = 2\left(-\frac{y}{2}\right) \Rightarrow x, -\frac{y}{2}, z$ are in A.P.

14. For $x = 1$, $y = a + b + c$
Tangent at $(1, a + b + c)$ is $\frac{1}{2}(y + a + b + c) = ax + \frac{b}{2}(x + 1) + c$
Comparing with $y = 2x$, $c = a$, $b = 2(1 - a)$

15. $x \in (2, 3) \Rightarrow -1 < x^2 - 4x + 3 < 0$, so f(x) is increasing in (–1, 0)
$\Rightarrow$ f(sin x) is increasing on $\bigcup_{n\in I}\left((4n-1)\frac{\pi}{2}, 2n\pi\right)$

16. Let point P on the straight line x + y = 4 be (m, 4 – m), this will be nearest to the parabola if $\perp$ at this point to the straight line becomes normal to the parabola.
Let it is normal at $x - 10 = t^2$, $y = 2t$
Perpendicular to x + y = 4 at (m, 4 – m) is $y - (4 - m) = (x - m)$ .....(1)
Normal at parabola at $(t^2 + 10, 2t)$ is $y + t(x - 10) = 12t + t^3$ .....(2)
(1) and (2) are same $\Rightarrow t = -1$, $m = \frac{17}{2}$ so required point is $\left(\frac{17}{2}, -\frac{9}{2}\right)$

17. $\left|\bar{a}\times\left(\bar{a}\times\bar{c}\right)\right| = \left|3\bar{b}\right| = 3\left|\bar{b}\right|$
$\left|\bar{a}\right|.\left|\left(\bar{a}\times\bar{c}\right)\right|\sin\frac{\pi}{2} = 3.1 \Rightarrow 3 = \left|\bar{a}\right|.\left(\left\|\bar{a}\right\| \left\|\bar{c}\right\|\sin\theta\right)$
$\Rightarrow 3 = 3.2\sin\theta \Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \cos^2\theta = \frac{3}{4}$

18. $\Delta'(x) = \begin{vmatrix} e^x & 2\cos 2x & 2x\sec^2 x^2 \\ \ln(1+x) & \cos x & \sin x \\ \cos x^2 & e^x - 1 & \sin x^2 \end{vmatrix} + \begin{vmatrix} e^x & \sin 2x & \tan x^2 \\ \frac{1}{(1+x)} & -\sin x & \cos x \\ \cos x^2 & e^x - 1 & \sin x^2 \end{vmatrix} + \begin{vmatrix} e^x & \sin 2x & \tan x^2 \\ \ln(1+x) & \cos x & \sin x \\ -2x\sin x^2 & e^x & 2x\cos x^2 \end{vmatrix}$
$= B + 2Cx + \ldots.$
Put x = 0, $B = \begin{vmatrix} 1 & 2 & 0 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 1 & 0 & -1 \\ 1 & 0 & 0 \end{vmatrix} + \begin{vmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{vmatrix} = 0$

19. Let g(x) be the inverse of f, then f(g(x)) = x
$\Rightarrow 3^{g(x)(g(x)-2)} = x$
$\Rightarrow (g(x))^2 - 2g(x) - \log_3 x = 0$
$\Rightarrow g(x) = \frac{2 \pm \sqrt{4 + 4\log_3 x}}{2} = 1 \pm \sqrt{1 + \log_3 x}$
Since $g : [1, \infty] \to [2, \infty]$
So $g(x) = 1 + \sqrt{1 + \log_3 x}$

20. Since $\sqrt{x^2 - 3x + 2} \geq 0 \Rightarrow 0 \leq \tan^{-1}\sqrt{x^2 - 3x + 2} < \frac{\pi}{2}$
Since $\sqrt{4x - x^2 - 3} \geq 0 \Rightarrow 0 < \cos^{-1}\sqrt{4x - x^2 - 3} \leq \frac{\pi}{2}$
$\Rightarrow 0 <$ L.H.S $< \pi \Rightarrow$ The given equation has no solution

21. Let $f = (5\sqrt{2} - 7)^{19}$
x – f = an integer $\Rightarrow [x] + \{x\} - f$ = an integer
$\Rightarrow \{x\} - f$ = an integer, but $-1 < \{x\} - f < 1 \Rightarrow \{x\} = f$
So, $x\{x\} = x.f = 1^{19} = 1$

23. $\frac{1}{x} = 2e\left[\frac{1}{3\ !} + \frac{2}{5\ !} + \frac{3}{7\ !} + \ldots\infty\right] = 2eS$

$S = \sum_{r=1}^{\infty} \frac{r}{(2\ \ r+1)!} = \frac{1}{2}\sum_{r=1}^{\infty} \frac{(2\ \ r+1)-1}{(2\ \ r+1)!} = \frac{1}{2}\sum_{r=1}^{\infty}\left[\frac{1}{(2\ \ r)!} - \frac{1}{(2\ \ r+1)!}\right]$

$= \frac{1}{2}\left[\frac{1}{2\ !} - \frac{1}{3\ !} + \frac{1}{4\ !} - \frac{1}{5\ !} + \ldots\infty\right]$

$= \frac{1}{2}e^{-1} = \frac{1}{2\ \ e}$

$\Rightarrow \frac{1}{x} = 2eS = \frac{2\ \ e}{2\ \ e} = 1$

So, $\int_0^1 f\ \ (y) \log_y x\, dy = 0$

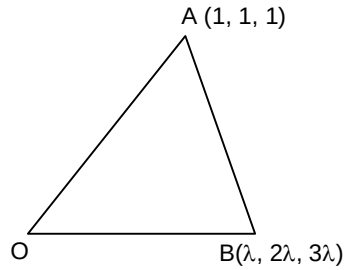
24. Let any point on second line be $(\lambda, 2\lambda, 3\lambda)$

$\cos\theta = \frac{6}{\sqrt{42}}$ $\sin\theta = \frac{6}{\sqrt{42}}$

$\Delta_{OAB} = \frac{1}{2}$ (OA).OB $\sin\theta = \frac{1}{2}\sqrt{3}\ .\lambda\sqrt{14} \times \frac{6}{\sqrt{42}} = \sqrt{6}$

$\Rightarrow \lambda = 2$

So, B is (2, 4, 6)

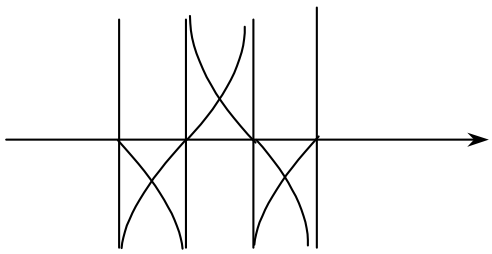


25. Centroid of triangle will be $\frac{\bar{a}+\bar{b}+\bar{c}}{3}$

Now line joining the orthocentre and the circumcentre is divided by centroid in 2 : 1 ratio internally, so orthocentre will be $\bar{a}+\bar{b}+\bar{c}$.

26. If we draw the graph of tan x and cot x, we observes that range of f (x) is $[-1, 0) \cup [1, \infty)$

So f (x) = $\frac{1}{\sqrt{3}}$ does not have any root.



27. $S - S_n < \frac{1}{300}$

$\Rightarrow \frac{1}{1-\frac{1}{3}} - \frac{1-\frac{1}{3^n}}{1-\frac{1}{3}} < \frac{1}{300} \Rightarrow \frac{3}{2}\left[1-1+\frac{1}{3^n}\right] < \frac{1}{300}$

$\Rightarrow \frac{1}{3^n} < \frac{1}{300} \times \frac{2}{3} \Rightarrow \frac{1}{3^n} < \frac{1}{450} \Rightarrow 3^n > 450$

$\therefore$ Least value of n = 6

28. $f(x) = \int_0^x e^{|t-x|}dt + \int_x^4 e^{|t-x|}dt = \int_0^x e^{x-t}dt + \int_x^4 e^{t-x}dt = -e^{x-t}\Big|_0^x + e^{t-x}\Big|_x^4 = e^x + e^{4-x} - 2$

$f'(x) = e^x - e^{4-x} = 0 \Rightarrow x = 4 - x \Rightarrow x = 2$

$f(0) = f(4) = e^4 - 1$, $f(2) = 2(e^2 - 1)$, so maximum value of $f(x)$ is $e^4 - 1$.

29. Given inequation can be rewritten as $3^{\sec^2 x}\sqrt{y^2 - \frac{2y}{3} + \frac{2}{9}} \le 1$

$\Rightarrow 3^{\sec^2 x}\sqrt{\left(y - \frac{1}{3}\right)^2 + \frac{1}{9}} \le 1$

Now, $3^{\sec^2 x} \ge 3$ and $\sqrt{\left(y - \frac{1}{3}\right)^2 + \frac{1}{9}} \ge \frac{1}{3}$

So, we should have $\sec^2 x = 1$, $y = \frac{1}{3}$

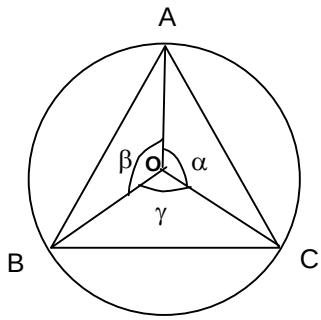
$\Rightarrow x = 0, \pi, 2\pi, 3\pi$

30. arc(AC) = 3, arc(AB) = 4, arc(BC) = 5

Let radius of the circle be 'r'

$\Rightarrow r\alpha = 3 \Rightarrow \alpha = 3/r$, $\beta = 4/r$, $\gamma = 5/r$

Now, $3 + 4 + 5 = 2\pi r \Rightarrow r = 6/\pi \Rightarrow \frac{1}{r} = \frac{\pi}{6}$



$\Delta_{ABC} = \Delta_{OAC} + \Delta_{OAB} + \Delta_{OBC}$

$= \frac{1}{2}r^2\left(\sin\left(\frac{3}{r}\right) + \sin\left(\frac{4}{r}\right) + \sin\left(\frac{5}{r}\right)\right)$

$= \frac{1}{2}.\frac{36}{\pi^2}\left(\sin\left(\frac{\pi}{2}\right) + \sin\left(\frac{2\pi}{3}\right) + \sin\left(\frac{5\pi}{6}\right)\right)$

$= \frac{18}{\pi^2}\left(1 + \frac{\sqrt{3}}{2} + \frac{1}{2}\right) = \frac{9\sqrt{3}\left(1+\sqrt{3}\right)}{\pi^2}$